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Total Number of Pages: 2

B.Tech
PCBM4302

5th Semester Regular / Back Examination 2016-17

SIGNALS AND SYSTEMS

BRANCH: AEIE, BIOTECH, ECE, EIE, ETC

Time: 3 Hours

Max Marks: 70

Q.CODE: Y289

**Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right hand margin indicate marks.**

Q1 Answer the following questions: (2 x 10)

- State whether the following discrete time system is static or dynamic where $y(n) = \log_{10}|x(n)|$
- State whether the following system is BIBO stable or not where $y(n) = x(n) - x(-n-1) + x(n-1)$
- Prove that $a_0 = 0$ and $a_n = 0$ (where a_0 and a_n represents trigonometric Fourier Series Coefficients) if $x(t)$ exhibits odd symmetry.
- Find the Fourier transform of A (constant).
- Find the Z transform of $x^*(n)$.
- Determine the energy and power of a d.t. unit step signal.
- Find the DFT of $\delta(n)$.
- Find the DFT of a purely real and even signal $x(n)$.
- Distinguish between FIR and IIR systems.
- Prove that autocorrelation at zero lag represents energy of a discrete time signal $x(n)$.

Q2 Define autocorrelation and cross correlation. Show that autocorrelation is an even function of lag 'l' and it is maximum at zero lag. (2+8)

Q3 a) Prove that the Parseval's relation in the Z domain (5)

$$\sum_{-\infty}^{\infty} x_1(n)x_2^*(n) = \frac{1}{2\pi j} \int X_1(Z)X_2^*\left(\frac{1}{Z^*}\right)Z^{-1}dZ$$

- b) Find the linear convolution between $x_1(n) * x_2(n)$ where (5)**
 $x_1(n) = \{1, 2, 3, 4\}$ and $x_2(n) = \{-1, 0, 1, 2\}$ using graphical method.

Q4 a) Perform a 4 point circular convolution using DFT/IDFT method where $x_1(n) = \{-1, 2, -3, 4\}$ and $x_2(n) = \{4, 7, 2\}$ (5)

b) Determine the inverse Z transform of $X(Z) = \ln \frac{\alpha}{\alpha - z^{-1}}$ ROC $|z| > \frac{1}{\alpha}$ (5)

Q5 a) Determine all possible values of $x(n)$ for $X(Z) = \frac{z^3 + z^2 + 1.5z + 0.5}{z^3 + 1.5z^2 + 0.5z}$ (5)

b) An LTI system is described by the difference equation $y(n] = x(n) + 0.81x(n-1) - 0.81x(n-2) - 0.45x(n-2)$. Determine the system function of the system. Sketch the poles and zeros on the Z plane. Access the stability. (5)

Q6 a) Solve the following difference equation using unilateral Z transform $y(n) - \frac{7}{12}y(n-1) + \frac{1}{12}y(n-2) = x(n)$ for $n \geq 0$ with initial conditions $y(-1) = 2$; $y(-2) = 4$ and $x(n) = \left(\frac{1}{5}\right)^n u(n)$ (5)

b) Define circular convolution. Derive the circular convolution relation between $x_1(n)$ and $x_2(n)$ (5)

Q7 Find the Fourier transform of periodic impulse train having amplitude 1 and time period T_0 . Find the trigonometric form of Fourier series of the half wave rectified sine wave having peak amplitude A (10)

Q8 Write short answer on any TWO: (5 x 2)

- Gibb's phenomena
- Sampling theorem
- Parseval's relation
- Stability analysis in the Z domain